

Baekseok Kim

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Skills

Mechanical Design: Mechanisms, actuators and passive joints, mechanical drawings, sensor/electronics integration, geometric-error analysis

CAD & Analysis: Creo, SolidWorks, CATIA, Inventor, Ansys, dynamics, motor torque-speed analysis

Prototyping & Test: Fabrication, assembly, FARO laser tracking, repeatability/payload/reliability testing

Robotics & Software: FK/IK, calibration, EtherCAT integration, motion control, Python, C/C++, MATLAB

Experience

Robotics Engineer / Project Manager

Dec 2018 – Feb 2022

Global Zeus (ZEUS Co., Ltd.) – Republic of Korea

- Led mechanical development of the company's first industrial delta robot, including actuators, passive joints and structural components, from CAD and analysis through fabrication and testing.
- Designed joint modules targeting IP67 and lubrication-free passive joints, plus a SCARA Z-roll module; developed FK/IK and calibration models to compensate for geometric errors.
- Reduced peak torque by more than 10% for the same test motion using motor torque-speed-aware trajectories; evaluated repeatability, payload and reliability with FARO measurement.

Research Assistant

Mar 2022 – Present

University of Nevada, Las Vegas – Las Vegas, NV

- Developed bipedal/humanoid platforms, including the Stilt-bot passive-ankle robot, integrating mechanical hardware with actuators, sensors and control software.
- Led whole-system research in humanoid loco-manipulation and tele-collaboration; served as lead control engineer for Team Avatar-Hubo, an ANA Avatar XPRIZE finalist.

Research Assistant

Mar 2016 – Feb 2019

Hanyang University – Republic of Korea

- Designed a skiing humanoid platform and MATLAB stability simulator, plus mechanisms for segmented rescue robots and interactive exhibition robots.

Selected Projects & Honors

- **Human-Powered Aircraft (2015):** SCOPE team leader; conceptual/3D design and aerodynamic/structural analysis, with responsibility for schedule, budget and resources.
- **Personal reduction mechanism (2019–2020):** 9:1 proof-of-concept prototype and 51:1 hollow-center design.
- **Recognition:** UKC 2023 Young Scholar Award; co-inventor of delta robot patent application KR20210042030A.

Education

University of Nevada, Las Vegas – PhD, Mechanical Engineering

Expected Dec 2026

Advisor: Paul Y. Oh; GPA: 4.0/4.0

Hanyang University – MS, Interdisciplinary Robot Engineering Systems

Feb 2019

GPA: 4.0/4.0

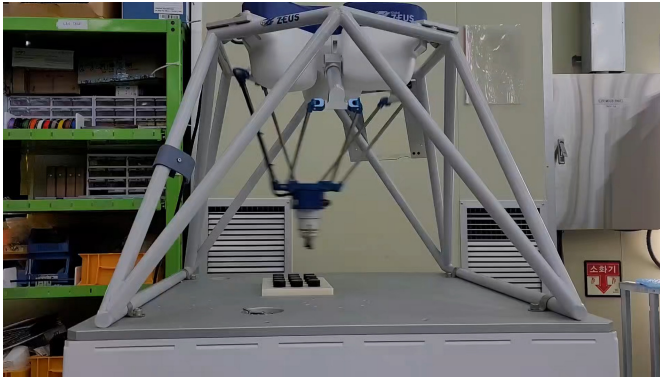
Sungkyunkwan University – BS, Mechanical Engineering

Feb 2017

Industrial delta robot: mechanical development

Global Zeus / 2019–2021

Actuator and passive-joint development for a modular industrial parallel robot.



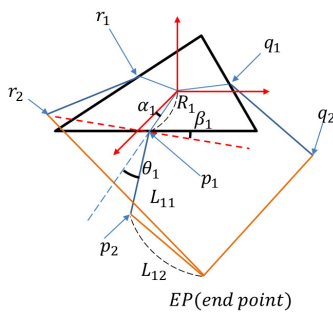
Industrial delta robot prototype



Roboworld 2020 exhibition

Mechanical design

- Led mechanical development of the company's first industrial parallel robot, including actuator modules, passive joints and load-bearing structures.
- Designed joint modules targeting IP67 and lubrication-free joints with two rotational degrees of freedom. Used Creo and Ansys through prototype fabrication, assembly and evaluation.



Kinematic model and geometric parameters



End-effector inspection from the development video

Prototype evaluation

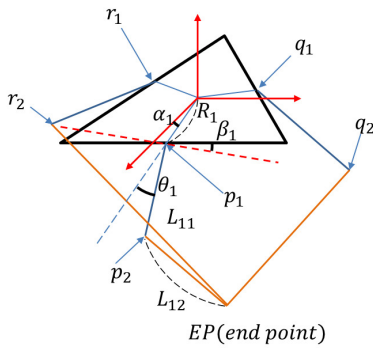
- Evaluated repeatability, payload and reliability using FARO laser-tracker measurement.
- Development footage includes actuator and joint fixtures, prototype operation and end-effector motion.

Project videos and full case study

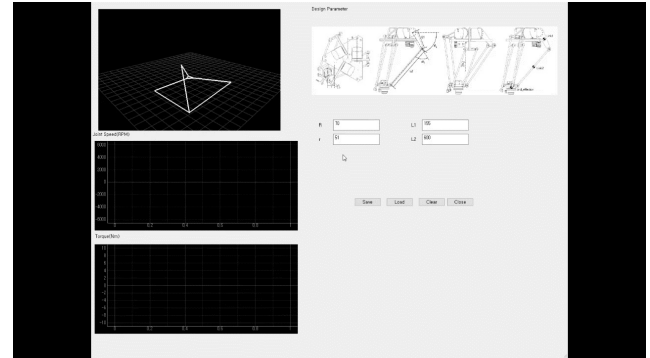
Industrial delta robot: modeling and validation

Global Zeus / 2019–2021

Geometric calibration, parameter sensitivity and dynamics used to support mechanical design.



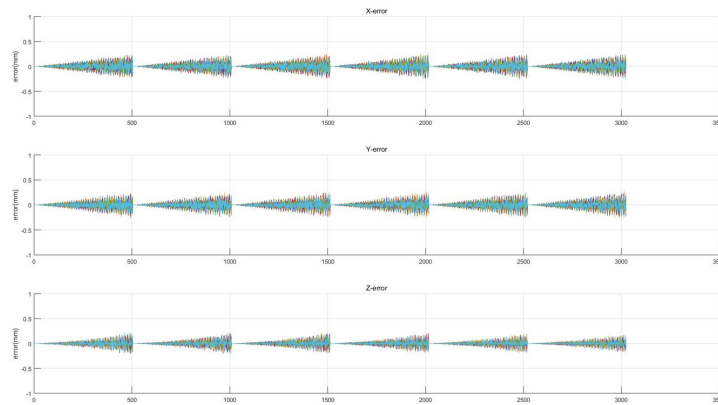
Kinematic model and geometric parameters



Dynamics simulator with speed and torque traces

Kinematics and dynamics

- Developed FK/IK models with geometric-error parameters for the modular mechanism.
- Developed a dynamics simulator to inspect speed and torque histories and support reliability analysis.



Position-error sensitivity study

Sensitivity analysis

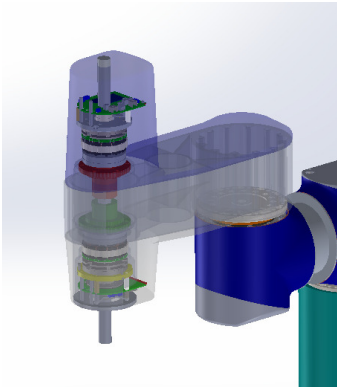
- Examined how geometric parameters affect end-point position error across the workspace.
- Compared effects of axis orientation, link geometry and base dimensions to inform design and calibration priorities.

Project videos and full case study

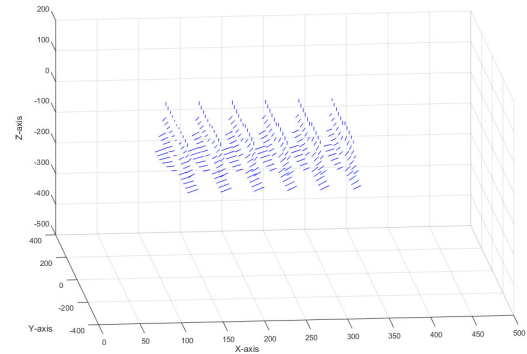
SCARA robot: Z-roll module and kinematics

Global Zeus / 2019–2021

Mechanical module development and explicit modeling of assembly-induced geometric error.



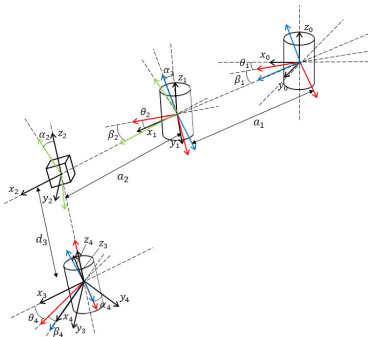
Z-roll module assembly



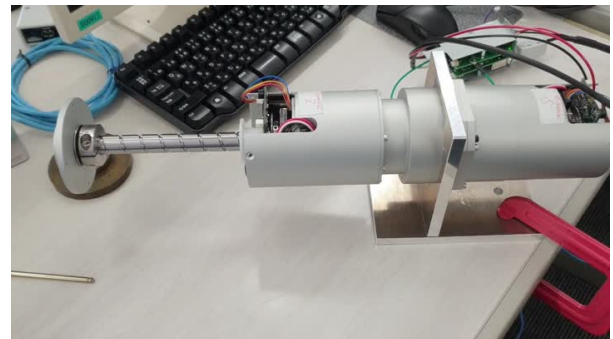
Predicted workspace position error

Z-roll module

- Designed and built the SCARA Z-roll module, integrating the linear and rotational mechanism with its actuator arrangement.



Kinematic model with manufacturing-error parameters



Physical Z-roll module on the test bench

Error-aware kinematics

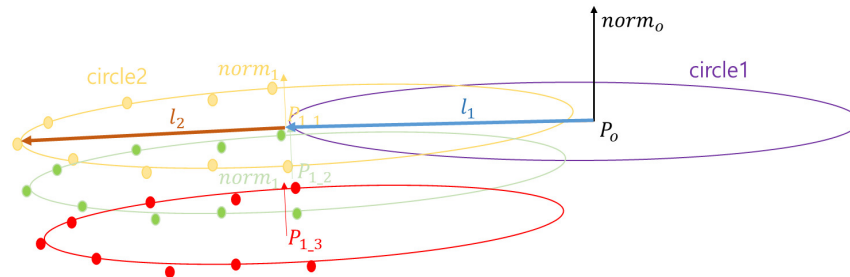
- Extended the model with manufacturing-error parameters, including an additional angular parameter beyond conventional DH variables.
- Developed forward and inverse kinematics to account for the resulting end-point position and orientation deviations.

Project videos and full case study

SCARA robot: calibration and motion optimization

Global Zeus / 2019–2021

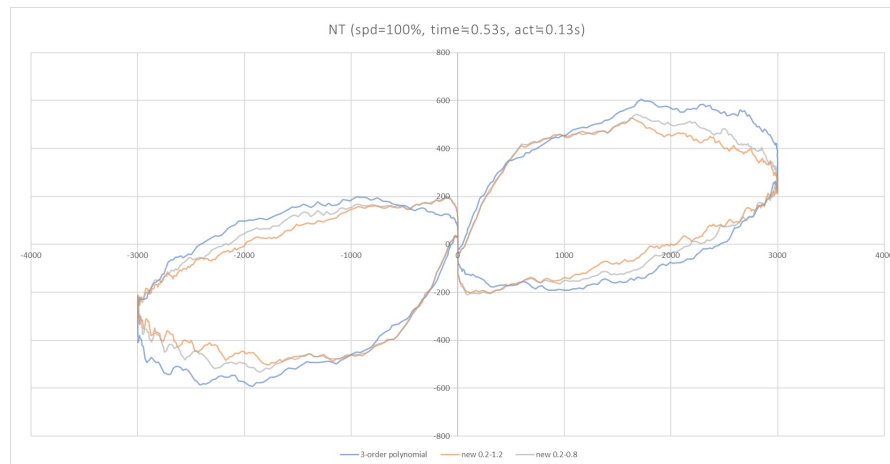
Geometry-based calibration and acceleration shaping based on motor torque-speed characteristics.



Geometry-based calibration model

Calibration method

- Used the mechanism's geometric constraints to estimate calibration parameters without deriving a separate Jacobian.
- Compared workspace behavior before and after calibration; the source test reports parameter estimation without a repeated calibration loop.



Comparison of motor torque and speed during the test motion

Motor loading

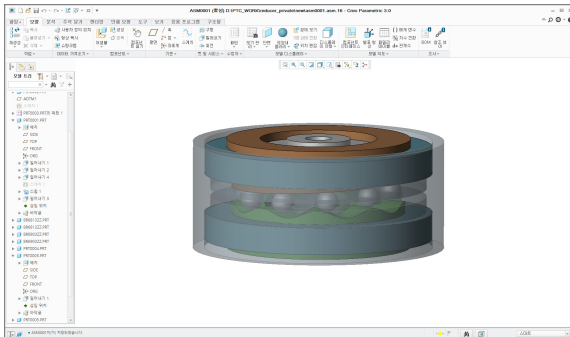
- Increased acceleration at low motor speed and reduced acceleration at high speed to use the available torque envelope.
- The original comparison reports more than 10% lower peak torque for the same test motion.

Project videos and full case study

Custom Reduction Mechanism

Personal project / 2019–2020

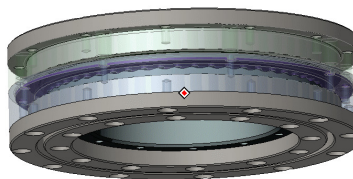
Ball-driven reduction mechanism: 9:1 prototype and a 51:1 hollow-center design.



Initial reduction mechanism CAD



Physical 9:1 prototype



51:1 hollow-center design

Operating principle

- An inclined input cam drives the balls axially. A retaining structure constrains each ball relative to the housing, while the profiled output surface produces reduced rotation.

Prototype development

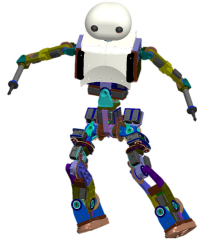
- Built a 9:1 prototype to check the operating principle. Developed a 51:1 version with a hollow center and examined the contact geometry in CAD.

Project videos and full case study

Humanoid Skiing Robot and Foot Sensor

Hanyang University / 2017–2018

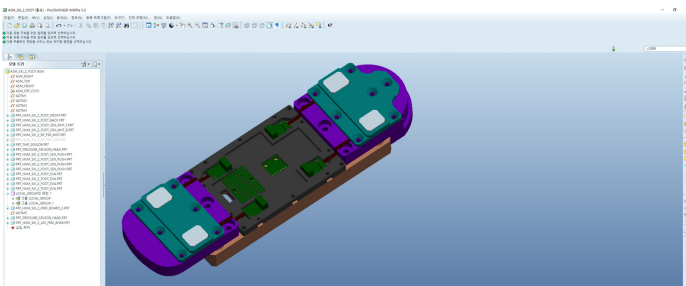
Humanoid mechanical design, foot-sensor hardware and dynamic stability simulation.



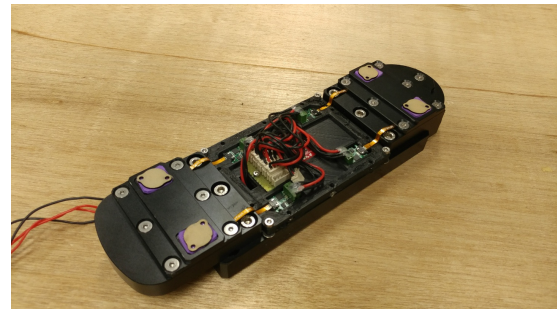
Humanoid robot CAD



Robot operating on a real ski slope



Foot-sensor assembly CAD



Fabricated foot-sensor hardware

Hardware development

- Designed the humanoid skiing platform in Creo and developed foot-sensor hardware, connecting mechanical design with sensing for stability analysis.

Dynamics and testing

- Developed a MATLAB dynamic simulator to study zero-moment-point behavior along ski trajectories. The wider project validated skiing on real slopes.
- Related publications include IEEE/RSJ IROS (2018) and Sensors 22(3), 816 (2022).

Project videos and full case study

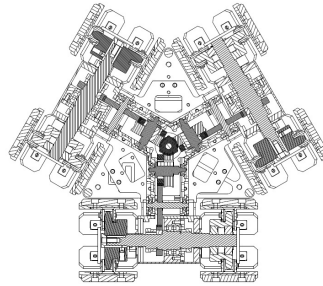
Segmented Disaster-Rescue Robot

Hanyang University / 2016–2017

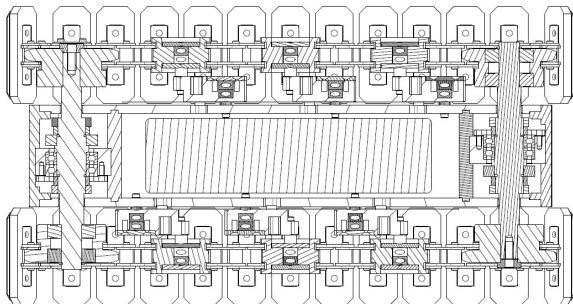
Tracked-module architecture, worm gearing and drivetrain packaging for disaster communication.



Segmented tracked-robot assembly



Module drivetrain cross section



Longitudinal drivetrain layout



Fabricated robot

Mechanical architecture

- Designed the mechanical structure of a segmented robot intended to support communication in disaster environments.

Drivetrain and assembly

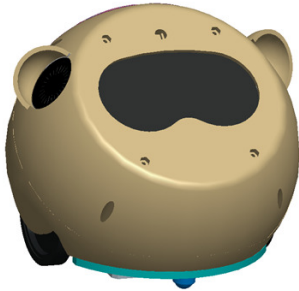
- Developed worm gearing, shafts and caterpillar-track sprockets within the module envelope.
- Designed bogie-wheel and chain-sprocket arrangements and progressed the CAD assemblies to physical hardware.

[Full project gallery](#)

EDIE Interactive Exhibition Robot

Hanyang University / 2016–2018

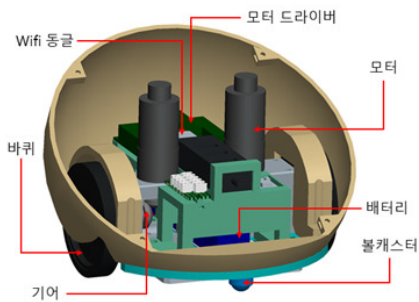
Compact enclosure design and packaging of drive, battery, sensing and display hardware.



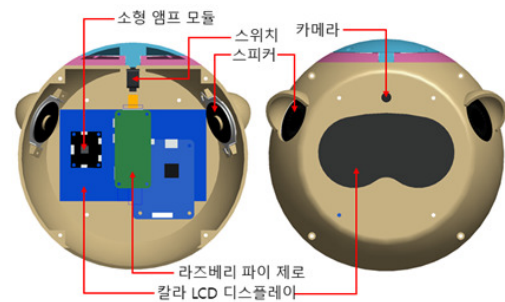
EDIE Ver. 3 CAD



Completed exhibition robot



Drive-system packaging



Camera, display and speaker packaging

Design and integration

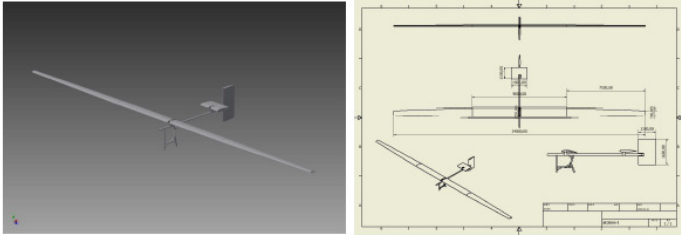
- Designed and fabricated mechanical structures for an interactive exhibition robot.
- Packaged motors, wheels, batteries and control electronics inside the character enclosure, with provision for the camera, display and speaker.

Project videos and full case study

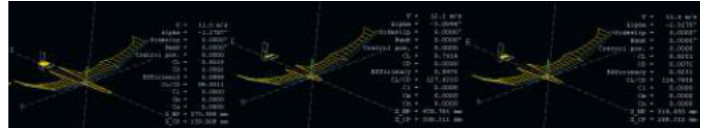
Human-Powered Aircraft

Sungkyunkwan University SCOPE / 2015

Team leadership, aircraft CAD, aerodynamic and structural analysis, and fabrication.



Aircraft CAD and engineering drawing



XFLR5 aircraft stability analysis



Wing fabrication



2015 competition

Program and design

- Led the SCOPE team, managing schedule, budget and resources for the 2015 Human-Powered Aircraft Competition.
- Created conceptual and 3D designs using Inventor and CATIA. Performed aerodynamic and structural analysis with Ansys and studied stability in XFLR5.

Fabrication and flight

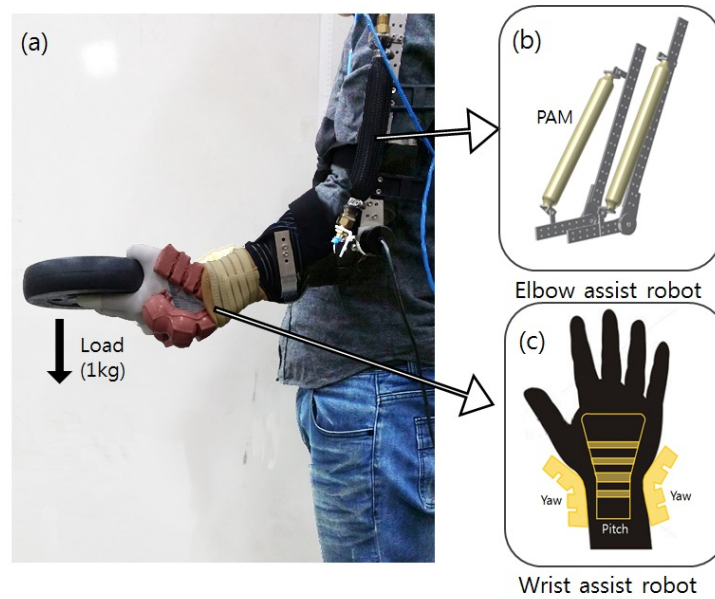
- Participated in the aircraft fabrication and assembly effort through competition flight activities.

Project videos and full case study

Soft-Actuator Assistive Robot

Sungkyunkwan University / 2015–2016

Early work on human-assistive mechanisms using soft actuation.



Elbow and wrist assistive-robot concepts

Design, fabrication and control

- Worked on the design, fabrication and control of an exoskeleton using soft actuators. This project formed the basis of undergraduate research on human-assistive robots.
- Related work: “Research of Human Assistive Robot using Soft Actuator,” KROC 2016.

[Full project gallery](#)